

Kéllia Umuhuza Gatete

Mechanical & Aerospace Engineer | Mechanical Design • Robotics Hardware • Test & Validation • R&D
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EDUCATION

Princeton University, Princeton, NJ

May 2026

B.S.E., Mechanical & Aerospace Engineering | Minor in Robotics — ABET-accredited programs in Mechanical and Aerospace Engineering.

Relevant Coursework: Engineering Design; Microprocessors for Measurement & Control; Autonomous Fabrication & Robotics; Materials Science; Fluid Dynamics; Rocket & Air-Breathing Propulsion

TECHNICAL SKILLS

Design & Manufacturing: Fusion 360, Rhino 8, Simplify3D/G-code, CAD/CAM, FEA, 3D printing, manual machining, materials selection, rapid prototyping

Testing & Instrumentation: MTS SENB/compression testing, rheometry, data acquisition, calibration and error analysis, thermal-resistance modelling, thermal cycling, dust/abrasion testing, pressure transducers, oscilloscopes

Robotics & Controls: Arduino/ATmega328P, Teensy, Raspberry Pi, ABB RobotStudio/RAPID/FlexPendant, SCARA, Hall-effect and temperature sensors, servos/relays, assembly programming, PID control

Software: Python, MATLAB/Simulink, C++, JavaScript, LabVIEW, Ansys STK/Astrogator, SPSS

ENGINEERING & RESEARCH EXPERIENCE

Senior Thesis Researcher — OSCAR MARSKIN, Princeton MAE

Sep 2025 – Apr 2026

Multimaterial soft-robotics protection and sensing system for Mars-inspired environments; co-authored 130-page thesis

- Co-designed and fabricated MARSKIN, a six-material flexible environmental skin for an origami crawler, combining TPU, EcoFlex, aerogel, PETG-PTFE, aluminium and a superhydrophobic coating; balanced insulation, structural support, adhesion, dust and abrasion protection, and compliant motion while moving from a circular test article to an OSCAR-specific panelized geometry compatible with folding.
- Validated MARSKIN against 2 controls using conductive/radiant heating, 30-min cold exposure, 3 hot-cold cycles, dust and abrasion: across 3 heat-lamp distances, maintained $\sim 24\text{--}27^\circ\text{C}$ peak through-thickness ΔT versus 4.6°C for controls; at 75°C conductive heating, improved peak/mean ΔT by $\sim 85\%$ with no visible delamination.
- Built the onboard sensing and data pipeline on a Raspberry Pi Zero 2 W, synchronising contact TMP117 and non-contact MLX90614 measurements at 1 Hz with 1080p day/night imaging every 10 s.

ACEE Summer Research Intern — Ju Lab, Princeton

Summer 2025

Combustion & Low Carbon Energy Conversion Laboratory | Prof. Yiguang Ju

- Investigated design-stage dielectric-barrier-discharge plasma conversion of CO_2 to graphene at $\sim 400^\circ\text{C}$ and evaluated 2 optical diagnostics — laser absorption spectroscopy and laser-induced fluorescence — for real-time CO/CO_2 monitoring and reactor optimisation.

Research & Gender Intern — Aegis Trust, Kigali, Rwanda

Summers 2023 & 2024

Two-year gender-norms programme with the University of Rwanda Centre for Gender Studies | Funded 2 $\times$ by the Summer Social Impact Internship Fund

- Analysed a 612-participant baseline study in SPSS across 5 districts and triangulated the quantitative results against document review, semi-structured interviews and a reflection workshop; the programme reached 3,753 stakeholders against a 2,600 target.
- Built assessment criteria for programme outcomes and co-developed an awards scheme with the Centre for Gender Studies, selecting 50 recipients through district committees — the deliverables that outlasted the funding cycle.

SELECTED ENGINEERING PROJECTS

LUREX Lunar Mission — Operations/Payload & CCD

Spring 2026

- Rebuilt a spacecraft onboard-computer budget after design-review feedback, replacing a percentage-based allocation with a throughput-driven method: sized workload in KIPS by software function across 5 operating modes and closed the governing case at 14.29 MHz against 1,000 MHz available; separately owned the mission risk matrix and cost compliance across 8 subsystems against a \$175M cap.

Robotic Additive Manufacturing — CEE 374

Fall 2025

- Ran SCARA and ABB IRB-6400/6700 6-DOF robotic AM workflows with G-code, RobotStudio/RAPID/FlexPendant, real-time height adjustment, and 1K/2K concrete extrusion; fabricated $20\times 23\times 75$ mm architected cement beams at 65% infill for 7-day MTS SENB fracture testing and evaluated 14-day topological columns under 2 mm/min compression.
- Characterised 3 cement pastes on a rheometer and selected the mix on buildability rather than extrudability (static yield 258 Pa, 20.7 mm maximum layer height against 10.8 mm), then printed $20\times 23\times 75$ mm Bouligand beams at 65% infill with 9° inter-layer rotation.
- Fracture-tested printed and cast beams in single-edge notch bending on an MTS Criterion at 0.05 mm/min: cast led on crack initiation (260 kN/m^{1.5}) but printed architectures inverted the ranking on energy-equivalent toughness (Bouligand 915 against cast 730), absorbing $\sim 70\%$ more energy per unit mass; traced the large Bouligand scatter to filament width, voids and interlayer bonding rather than to the architecture.

Automated Railway Control System — MAE 412

Spring 2025

- Built a hands-free unload/sort/reload platform using 3 Hall-effect sensors, 2 rail switches, 2 servos, 1 relay and 3D-printed mechanisms to handle 4 stainless-steel balls across 2 sizes; achieved all objectives plus the stretch reloading goal with repeatable class-demonstration operation.
- Implemented Arduino $\leftrightarrow$ ACIA serial and assembly sequencing with host control, coordinating sensor-triggered routing, speed transitions, track-power stops, unloading and parity-based payload reloading.

Hydrogen-Electric Hybrid Aircraft Concept — MAE 426

Spring 2025

- Co-developed a parallel hydrogen-electric concept benchmarked against the 210-passenger Boeing 737-8; combined direct-injection H_2 ICE, SOFC and PMSM power paths in a 30/70 split to estimate 53.5% idealized peak system efficiency, exceeding the project's 50% target.

HONOURS, FELLOWSHIPS & FUNDING

John Marshall II Memorial Senior Thesis Fund — 2026 | Competitive proposal funding for OSCAR MARSKIN.

ACEE Summer Internship Program / Peter B. Lewis Fund — Summer 2025 | Competitive funded energy and environment research internship.

Summer Social Impact Internship (SSII) Fund — Recipient, 2 $\times$

Academic Excellence (Rwanda) — First Lady Academic Excellence Award, 2018 & 2022; Rwanda Education Board Award, 2021 (2nd nationwide, A-Level, Gashora Girls Academy of Science & Technology); AIMS Outstanding Achievement Award, 2019 (4th nationally, Mathematics).

Languages — English, French, Kinyarwanda, Kirundi, Kiswahili, Spanish, German